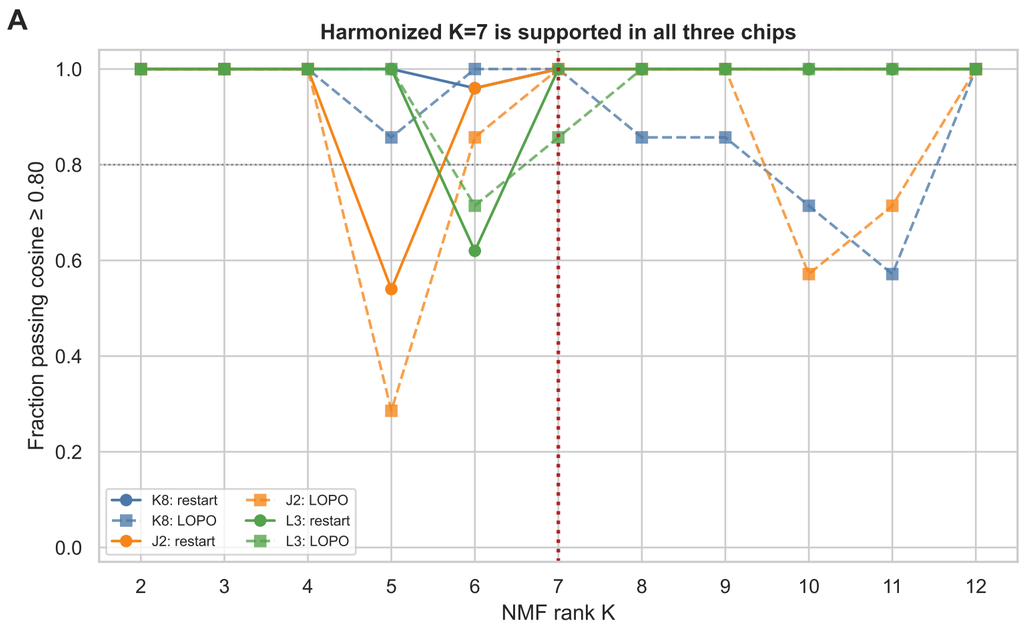
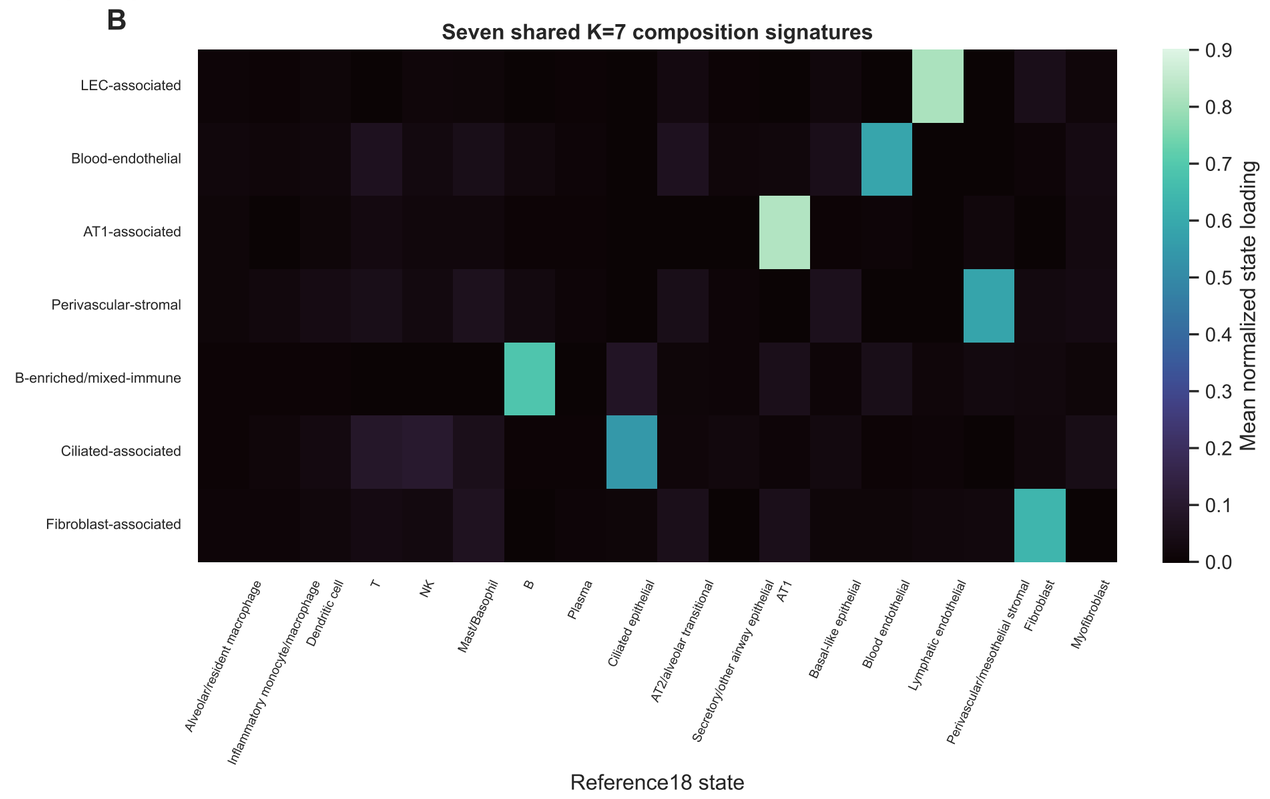
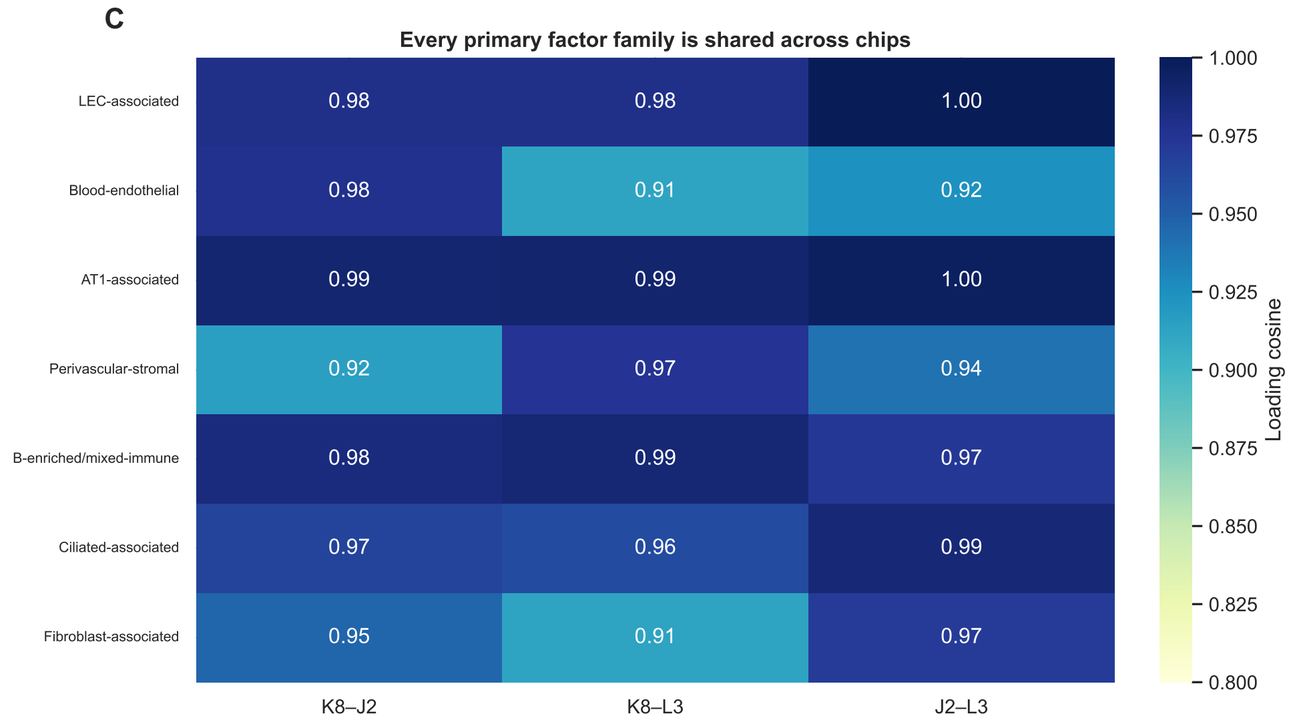
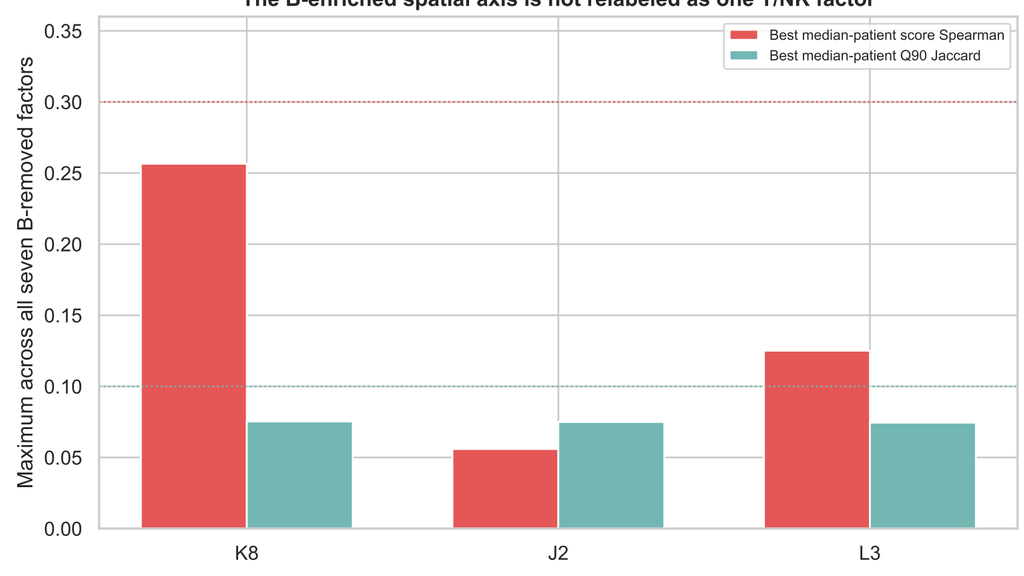
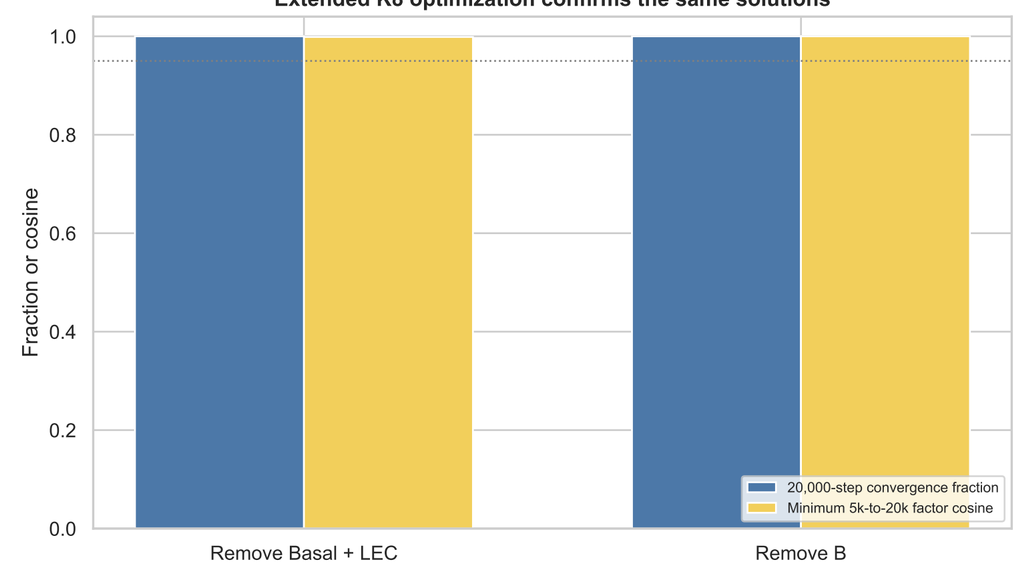
Harmonized K=7 Neighborhood NMF is cross-chip reproducible and state-removal robust











Primary model: frozen 18-state, centre-excluded 50-µm Neighborhood NMF; 50 restarts; 495 balanced observations per patient; disease labels unused. B is interpreted as B-enriched/mixed-immune; LEC remains sensitivity-qualified.